

Exp. 4	PROVISION A VPS INSTANCE FROM A CLOUD PROVIDER OR SET UP A VIRTUAL MACHINE ON YOUR LOCAL SYSTEM USING SOFTWARE LIKE VIRTUALBOX OR VMWARE
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AIM

To provision a Virtual Private Server (VPS) using VirtualBox, install Ubuntu Server, configure network settings (IP Address, Subnet Mask, Gateway, DNS), and enable SSH remote access.

OBJECTIVES

1. To understand the concept of Virtual Private Server (VPS).
2. To create and manage virtual machines using VirtualBox.
3. To install Ubuntu Server OS.
4. To configure static network settings.
5. To enable SSH remote access.

SOFTWARE REQUIREMENTS

- Oracle VirtualBox
- Ubuntu Server 22.04 LTS ISO
- Windows/Linux Host System
- Internet Connection
- Terminal Access

THEORY

A Virtual Private Server (VPS) is a software-based server created using virtualization technology. It operates independently with its own operating system, storage, memory, and network configuration while sharing the physical hardware with other virtual machines.

Virtualization allows efficient utilization of resources and forms the foundation of cloud computing platforms.

Network configuration parameters include:

IP ADDRESS

A unique address assigned to a device on a network.

SUBNET MASK

Determines the network and host portions of an IP address.

GATEWAY

Provides access to networks outside the local network.

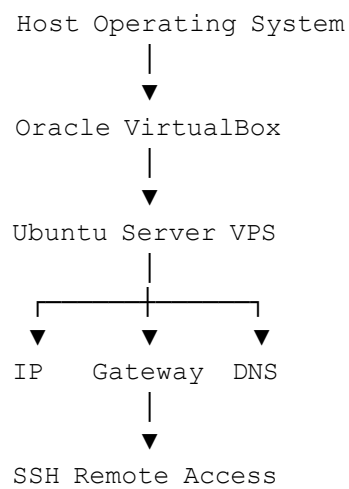
DNS (DOMAIN NAME SYSTEM)

Converts domain names into IP addresses.

SSH (SECURE SHELL)

A secure protocol used for remote administration of Linux systems.

ARCHITECTURE



ALGORITHM

1. Install Oracle VirtualBox.
2. Create a new virtual machine.
3. Attach Ubuntu Server ISO image.
4. Install Ubuntu Server.
5. Update system packages.
6. Identify network interface.
7. Configure static IP address.
8. Configure subnet mask.
9. Configure default gateway.
10. Configure DNS server.

PROCEDURE

STEP 1: INSTALL VIRTUALBOX

Download and install Oracle VirtualBox.

STEP 2: CREATE VIRTUAL

MACHINE

Configure:

Parameter	Value
Name	VPS-Server
OS Type	Ubuntu (64-bit)
RAM	2 GB
CPU	2 Cores
Storage	20 GB

STEP 3: INSTALL UBUNTU SERVER

1. Attach Ubuntu Server ISO.
2. Start VM.
3. Follow installation wizard.
4. Create username and password.
5. Complete installation.

STEP 4: UPDATE UBUNTU

```
sudo apt update  
sudo apt upgrade -y
```

STEP 5: IDENTIFY NETWORK INTERFACE

```
ip a
```

Sample Interface:

```
ens33
```

STEP 6: CONFIGURE STATIC IP

Edit Netplan file:

```
sudo nano /etc/netplan/00-installer-config.yaml
```

Add:

```
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.1.100/24
      gateway4: 192.168.1.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
```

Apply configuration:

```
sudo netplan apply
```

STEP 7: VERIFY NETWORK CONFIGURATION

Check IP:

```
ip addr
```

Check routing table:

```
ip route
```

Check DNS:

```
cat /etc/resolv.conf
```

STEP 8: TEST CONNECTIVITY

Test Internet:

```
ping -c 4 8.8.8.8
```

Test DNS:

```
ping -c 4 google.com
```

STEP 9: INSTALL SSH SERVER

```
sudo apt install openssh-server -y
```

STEP 10: ENABLE SSH SERVICE

```
sudo systemctl enable ssh
sudo systemctl start ssh
```

Verify status:

```
sudo systemctl status ssh
```

STEP 11: CONNECT REMOTELY

From another system:

```
ssh username@192.168.1.100
```

COMMANDS USED

```
sudo apt update
sudo apt upgrade -y
ip a
sudo nano /etc/netplan/00-installer-config.yaml
sudo netplan apply
ip addr
ip route
cat /etc/resolv.conf
ping -c 4 8.8.8.8
ping -c 4 google.com
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh
ssh username@192.168.1.100
```

SAMPLE OUTPUT

IP CONFIGURATION

```
inet 192.168.1.100/24
```

ROUTING TABLE

```
default via 192.168.1.1 dev ens33
```

DNS CONFIGURATION

```
nameserver 8.8.8.8
nameserver 8.8.4.4
```

OBSERVATIONS

Parameter	Status
VM Creation	Successful
Ubuntu Installation	Successful
Static IP Configuration	Successful
Subnet Mask Configuration	Successful
Gateway Configuration	Successful
DNS Configuration	Successful
Internet Connectivity	Verified
SSH Installation	Successful
SSH Service Running	Verified
Remote Login	Successful

RESULT

A Virtual Private Server (VPS) was successfully created using Oracle VirtualBox. Ubuntu Server was installed and configured with static IP address, subnet mask, gateway, and DNS settings. SSH service was installed, enabled, and verified for secure remote access.